GIS-Based Land Suitability Analysis for Urban Development using Land Use Land Cover

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October 2026

Live demo: `iamabhinav-dev.github.io/gis-urban-suitability`

CS60111 Geographical Information Systems

Problem and approach

Where can new development reasonably go?

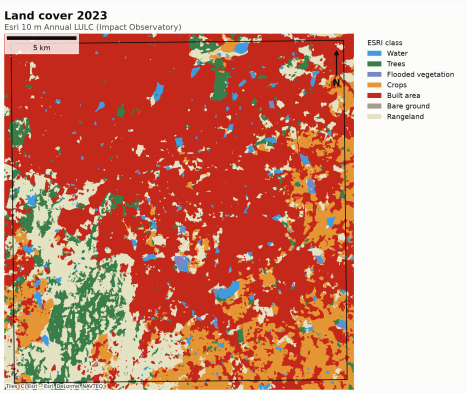
- Classic GIS answer: weighted overlay with **expert (AHP) weights**, rarely checked against reality.
- Ours: **learn from where building actually happened**, and test every method against **real growth**.

Three questions

1. Which land is **built-up**, **forest** or **usable**?
2. Given a building at **A**, where are similar locations **B**?
3. Do data-driven scores beat a classic MCDA overlay?

All in **Python**; 2018 data only → tested on growth by 2022/23.

Study area and data



Bengaluru South: 582 km^2, 58,218 cells of 100 m

- Hosur Road corridor, Kanakapura Road, Bannerghatta NP
- 8.4 % of non-built land became **persistently** built by 2022/23

Open data

- ESRI 10 m LULC 2018–2023 (WorldCover 2021: cross-check)
- Copernicus DEM; OSM roads, water, buildings
- OSM **as of 1 Jan 2018** for the test: today's roads are 2.2× longer

Validation design and leakage rules

Two runs

- **Validation run**: only data up to 2019 (LULC 2018, OSM 2018-01-01) → tested on growth by 2022/23
- **Final run**: same models on 2023 data → the map

Persistent growth: non-built in 2018 *and* 2019, built in 2022 *and* 2023

12,777 candidate cells, **1,125 grew** (8.8 %)

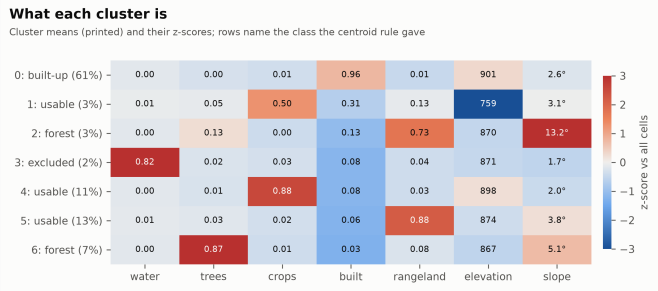
Leakage rules

- no model sees the cell's own land cover
- nothing after 2019 feeds a 2018 score
- trained models scored **out of fold** (2 km blocks)

Always shown: random and distance to built-up (the bar to beat)

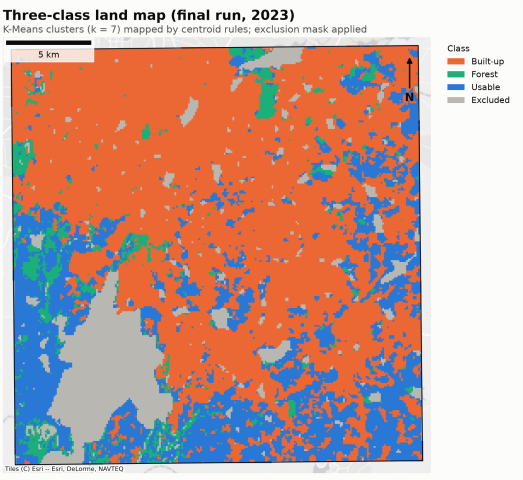
Metrics: AUC, lift in the top 10 %, TOC, adjacent vs outlying growth

Land classification: clustering



- K-Means on 5 land-cover shares + elevation + slope
- **k = 7**: best silhouette (0.61); GMM worse at every k
- Two extra clusters: **steep scrub forest** (Bannerghatta hills, 13°) and the **low western valley**
- Centroid rules → classes; steep natural vegetation = forest

Land classification: three-class map



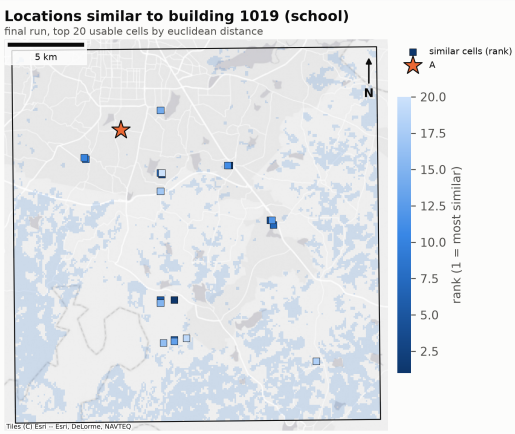
2018 → 2023

- **Built-up** 306 → **348** km^2
- **Usable** 182 → 138 km^2
- **Forest** 29 km^2; excluded 66 km^2

Agreement on land cells

- ESRI: **96.6 %**
- WorldCover: 64 %; the products disagree (built-up 62 % vs 33 %)

Building similarity: A → B

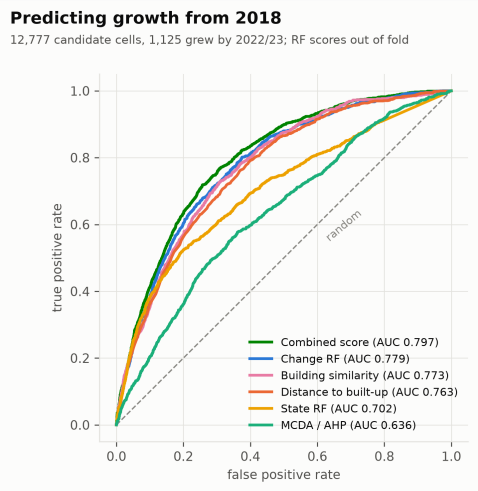


Profile: land cover in 250 / 500 m rings, distances to built-up, roads and water, road density, terrain (never the cell itself)

- Query a building → top 20 **usable** cells by Euclidean distance
- Each match explained: closest and most different features
- Self-check: a building's own location is always in the top 1 %

Aggregate: mean distance to the 10 nearest building profiles
→ **AUC 0.773** on growth, without ever seeing a growth label

Learned models: predicting growth from 2018



Change RF: 2018 surroundings of the 355 cells built by 2020/21

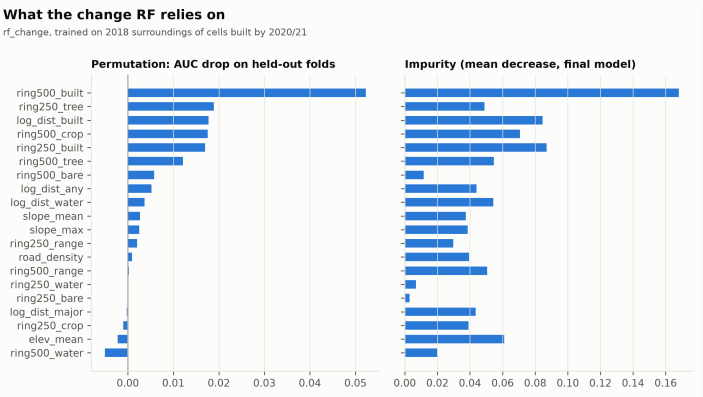
- **AUC 0.779**, above distance to built-up (0.763)
- In-sample it would claim 0.997 → **out-of-fold** scores are essential

State RF (built vs non-built): 0.998 on its own task, only 0.702 on growth

Held-out buildings (2 km blocks): RF finds unseen building cells with AUC 0.98

Combined with the similarity: **AUC 0.797**

What the change RF relies on



Built-up land within 500 m dominates; trees, crops and distance to built-up follow; terrain and water add little.

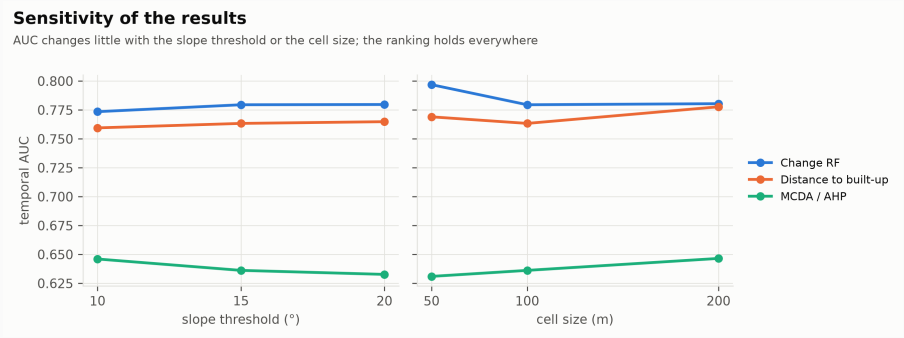
The classic baseline: MCDA / AHP

Five criteria, AHP weights, **consistency ratio 0.003**

Criterion	Weight	AUC alone
Distance to built-up	0.30	**0.763**
Distance to major road	0.30	0.548
Slope	0.16	0.612
Land cover	0.16	0.527
Water setback	0.09	0.518
Overlay		**0.636**

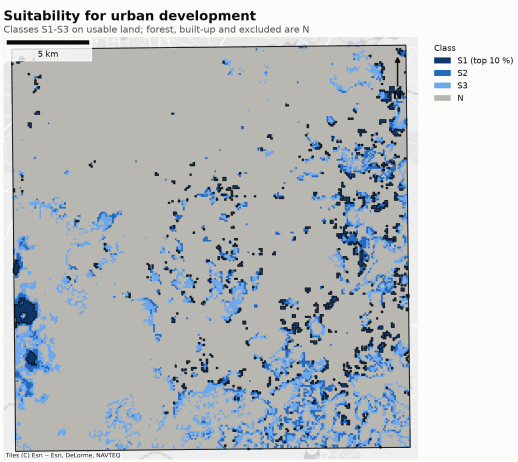
- The overlay does **worse than one of its own criteria**
- 70 % of the weight sits on near-random criteria and dilutes the one strong signal
- Weights are not tuned on the test label: this is what fixed expert weights do

Sensitivity



Slope threshold, k and cell size barely matter (AUC within 0.02), and the **ranking holds everywhere**: change RF > distance to built-up > MCDA.

Suitability map and candidate sites

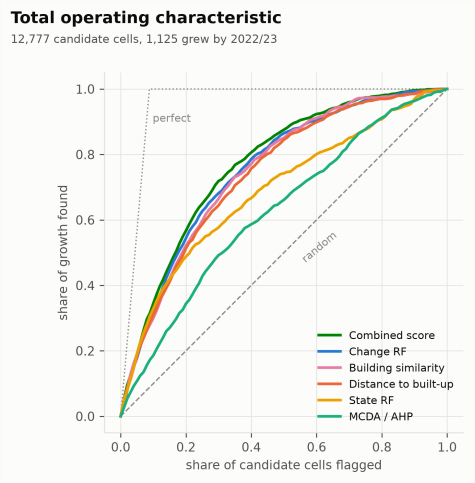


Combined score $= 0.3 \times$ similarity $+ 0.7 \times$ change RF (weight chosen on the 2020/21 label, not the test label)

- Classes on usable land: S1 top 10 %, S2, S3, N
- S1 cells merged into **557 patches** (1,382 ha)
- **Top 20 sites** ($\geq$ 5 ha): flat, next to built-up land, rangeland or crops; none in a lake or the national park

Interactive map with all layers: …/`map.html` (live demo)

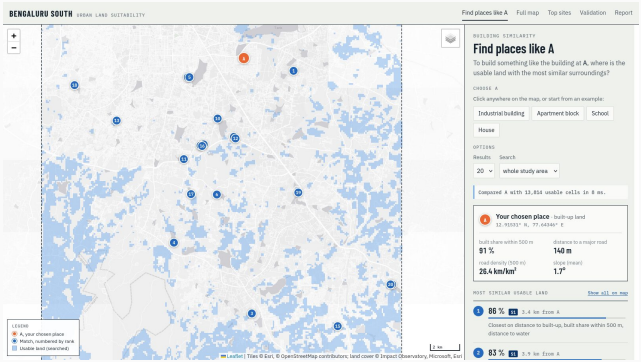
Validation: every score against real growth



Score	AUC	Lift top 10 %
Combined	**0.797**	**3.45**
Change RF	0.779	3.23
Similarity	0.773	2.99
Dist. to built-up	0.763	3.04
MCDA / AHP	0.636	1.86
Random	0.50	1.00

- S1 + S2 hold **65 %** of the growth
- Adjacent growth AUC 0.82; **outlying growth 0.51** (chance)
- Built-up nearby carries the signal; roads and water add nothing

Live demo



`iamabhinav-dev.github.io/gis-urban-suitability`

- **Find places like A**: click any building → the 20 most similar **usable** cells, each explained feature by feature
- Same answer as the Python query (checked on 30 buildings)
- Suitability map with the **top 20 sites**, the full interactive map, report and slides; the query runs in the browser

Limitations

- Growth here is mostly **edge expansion**: learned models gain only a little over "next to existing built-up land".
- **OSM completeness**: road mapping more than doubled since 2018; OSM buildings cover $\sim$1 % of built cells in the south (2018); almost all untyped.
- **Land-cover errors**: ESRI generalises urban areas and calls scrub forest "rangeland"; growth labels inherit its errors.
- **Past growth $\neq$ good planning**: zoning, flood risk and infrastructure capacity are not in the data.

Conclusion

1. **Land classes**: K-Means ($k = 7$) $\rightarrow$ **built-up** 348, **forest** 29, **usable** 138 km^2 (2023); 96.6 % agreement with ESRI
2. **Building similarity**: any building $\rightarrow$ similar usable locations, explained; predicts growth (AUC 0.773) without labels
3. **Suitability**: the combined score (**AUC 0.797**, lift 3.45) beats distance to built-up (0.763) and a classic AHP overlay (0.636)

Lesson: honest validation (leakage rules, baselines) matters more than the model: most of the signal is "next to existing built-up land"; growth away from it is not predicted.

Thank you

Questions?

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